

PROJECT ABSTRACT[Form-1] (YEAR-2026-27)

PROJECT ID: SKIT/CSE/2023-2027/26 ^{CSE-150} BRANCH: Computer Science & Engineering SECTION: B.

TITLE OF PROJECT: SecureCred - Blockchain Based Digital Certification System

SDG MAPPING: SDG 4 (Write SDG Title) Quality Education

PROJECT TRACK: (Tick the appropriate one / ones)

R&D INNOVATION ☒	CONSULTANCY PROJECTS	DEPARTMENT AS CLIENT PROJECTS	CLIENT PROJECT	HARDWARE-BASED PROJECTS
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MANDATORY EXTERNAL EVALUATION OF PROJECT (Tick the appropriate one)

RESEARCH PAPER PUBLICATION ☒	PATENT FILING / GRANT	POSTER COMPETITION
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PROBLEM STATEMENT:

Existing academic and vocational certification system rely on paper certificates or plain PDF/centralized records, which are vulnerable to forgery and tampering and offer no cryptographic proof of authenticity. Verifying a certificate typically require manually contacting the issuing institute, which is slow and unreliable, and leaves no audit trail for issuance or revocation. Learners can also permanently lose access to their credentials if an institute shuts down or misplace its record. SecureCred securely issue and verifies vocational credentials by cryptographically hashing (SHA-256) each certificate, storing it via decentralized DDFS.

NAME OF TOOL / TECHNOLOGY	FRONTEND/ BACKEND	SOFTWARE/ HARDWARE	PURPOSE OF USE
React.js	Frontend	Software	Build responsive user interface and dashboard
Tailwind CSS	Frontend	Software	Utility - first styling for responsive UI design
Figma	Frontend	Software	UI/UX prototype and interface design

Node.js	Backend	Software	server-side javascript runtime environment
Express.js	Backend	Software	REST API development and backend routing.
Convex	Backend	Software	Backend application and data service
PostgreSQL	Database	Software	store-off-chain certificate metadata
IPFS	Storage	Software	Decentralized storage for certificate files
Pinata	Storage	Software	IPFS pinning service for certificate reliable files
SHA-256	Security	Software	Generate Cryptographic hash of each certificate
Solidity	Blockchain	Software	Develop Smart Contracts
Polygon	Blockchain	Software	Anchor certificate hash & CID on-chain
Ganache	Blockchain	Software	local blockchain Environment
Docker	Deployment	Software	Containerize application for deployment
Git	Deployment	Software	version control and team collaboration

PROPOSED PROJECT SPRINTS (add more row if required):

S.No.	SPRINT NAME	PROPOSED USER STORY OF RESPECTIVE SPRINT
1	System Architecture and Setup	we want to design the system architecture, set up the Github repositories, backend/ frontend project scaffolding, and the PostgreSQL database schema so that the technical foundation for parallel development is ready
2	Core Module Development	we want to develop the core smart contract backend REST APIs and frontend UI/ dashboard components in parallel so that the primary certificate issuance module are functional.

3	Feature Integration & Verification	we want to implement certificate verification and revocation logic, authentication IPFS/Pinata storage integration and the verifier portal so that end-to-end issuance and verification work.
4	Backend Blockchain Frontend Integration	we want to integrate the backend - blockchain and frontend layers and deploy smart contract to a public testnet so that the complete workflow operate across all modules
5	Security Hardening and Testing	we want to conduct smart contract security audits, API security hardening, and end-to-end integration testing so that the system is secure and reliable before release.
6	Final Deployment and Release	we want to complete regression testing, finalize production deployment on main-net / Cloud infrastructure testing so that the system is secure and reliable before release.

TEAM MEMBER DETAILS:

STUDENT NAME	MOBILE No.	EXPERTISE AREA	ROLE IN PROJECT
Kartik Marwal	8386847113	Blockchain, Backend	Team Lead, Blockchain and System Architecture
Jaideep Singh Gehauhan	8949567353	Backend, Database	Backend API, PostgreSQL, Authentication
Keshav Bheel	7850947034	Frontend	UI Development, React, Dashboard Design
Karish Mangal	8435057127	Testing	Testing, Documentation Security.

Verified & Approved By:
(MENTOR'S NAME & SIGNATURE)
Project Mentor

Vikram
Khandelwal

July
22/6/24

ROLES & RESPONSIBILITY OF TEAM MEMBERS [Form - 2]

CSE-150

PROJECT ID: SKIT/CSE/2023-2027/.. BRANCH: Computer Science & Engineering SECTION: B

TITLE OF PROJECT: SecureCred Blockchain Based Digital Certification System

TEAM LEAD: Kartik Marwal

SPRINT(s) NAME: 1) System Architecture and Setup

2) Feature Integration and verification

3) Backend blockchain frontend integration

USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
System Architecture and Blockchain Design	10/08/2026	22/08/2026	Design the system architecture blockchain workflow and certificate lifecycle connecting frontend backend, IPFS and Polygon
Smart Contract Development Issuance & Storage	23/08/2026	3/10/2026	Develop the solidity smart contract for certificate issuance and hash storage; deploy and test on Ganache.
Verification & Revocation Contract logic	4/10/2026	31/10/2026	Implement on-chain certificate hash verification and issuer controlled revocation logic
Polygon Testnet Deployment & Gas Optimization	1/11/2026	2/01/2027	Deploy smart contract to the polygon amoy testnet, optimize gas usage, and integrate contract events with backend listeners.
Smart Contract Security Audit & Hardening	3/01/2027	06/02/2027	Conduct a security audit of the smart contracts and patch identified vulnerabilities
Mainnet Deployment & Final Blockchain Integration.	07/02/2027	15/03/2027	Finalize contract deployment on the polygon product networks and complete end-to-end blockchain integration testing.

MEMBER 1 Jaideep singh
chaubhan

SPRINT(S) NAME 1) Core Module Development
2) Feature integration and
Verification
3) Backend Blockchain Frontend
Integration

USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
Backend setup & DB schema Design	10/08/2026	22/08/2026	Set up Node.js/Express Backend project structure and design the PostgreSQL database schema for certificate metadata storage
Certificate REST API Development	23/08/2026	03/10/2026	Develop REST API endpoints for certificate creation, retrieval and metadata management
Authentication Module & IPFS / Pinata Integration	04/10/2026	31/10/2026	Implement JWT-based authentication with role-based access control, and integrate the Pinata SDK to upload certificate file to IPFS.
Backend - Blockchain Integration	01/11/2026	02/1/2027	Connect Backend APIs with smart contract functions for issuance, verification and revocation, including transaction status tracking.
API security Hardening & Rate Limiting	03/01/2027	06/02/2027	Implement input validation, rate limiting, and fix vulnerabilities identified during security testing
Backend Deployment and Production Readiness	07/04/2027	15/03/2027	Containerize the backend with Docker, configure the production database, and finalize deployment scripts

MEMBER 2: Keshav Bheel

SPRINT(S) NAME: 1) Backend Blockchain
Frontend integration.

- 2) Feature integration and verification
- 3) Final Deployment & release

USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
UI Component library Development	10/08/2026	21/08/2026	Build reusable React.js UI component and convert Figma layouts into functional code using Tailwind CSS.
Institution & Student Dashboard development	23/08/2026	03/10/2026	Develop institution dashboard for certificate issuance / management and the student dashboard for viewing issued certificate.
Verifier Portal - QR/ID Verification UI	04/10/2026	31/10/2026	Build the verifier portal with QR code scanning and ID-based certificate verification flow.
Frontend-Backend & Blockchain API Integration	01/11/2026	02/01/2027	Integrate frontend dashboard with backend REST APIs and blockchain transaction status for live certification.
UI/UX Refinement & Bug Fixing	03/01/2027	06/02/2027	Refine UI/UX based on testing feedback and fix bugs identified during integration testing.
Frontend Production Deployment	07/02/2027	15/03/2027	Containerize and deploy the frontend application to production using Docker.

MEMBER 3: Kavish Mangal

SPRINT(S) NAME:

- 1) Security Hardening & Testing
- 2) Final deployment and release

USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
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Test Automation Framework Setup	10/08/2026	22/08/2026	Set up the Hardhat / chai test framework and Ganache Local Blockchain environment for automated Smart contract testing
Smart Contract Unit test development	23/08/2026	03/10/2026	Write automated unit test scripts covering certificate issuance, verification and revocation contract functions (40% milestone)
Backend API test Automation	04/10/2026	31/10/2026	Develop automated test scripts (Postman / Test) for backend REST API endpoints (50% milestone)
Security testing - Tamper & Access control.	01/11/2026	02/01/2027	Perform security testing for certificate hash - tampering detection and role based access control vulnerabilities (70% milestone)
End-to-end Integration Testing	03/01/2027	06/02/2027	Conduct integration testing across frontend, backend and Blockchain components to validate complete certification issuance and verification flow.
Regression Testing and Release	07/02/2027	15/03/2027	Fix identified bugs, execute regression testing, and validate system readiness for final release deployment (100% milestone)

Verified & Approved By:
(MENTOR'S NAME & SIGNATURE)
Project Mentor

Vikram
Khandelwal

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22/8/26